

Elias Dunham

Boston, MA | (610) 972-8554 | dunham.eli@northeastern.edu | linkedin.com/in/elias-dunham | [Portfolio](#)

EDUCATION

Northeastern University, College of Engineering

Boston, MA

Candidate for Bachelor of Science in Mechanical Engineering and Physics

May 2028

GPA: 3.8

Coursework: Fluid Mechanics, Thermodynamics, Mechanics of Materials, Materials Science, Differential Equations, Physics 2

Honors & Extracurriculars: Dean's List, American Society of Mechanical Engineers, Northeastern Composition Club, Delta Tau Delta

Northeastern University, Dialogue of Civilizations

Panama City, Panama

International Applications of Fluid Mechanics

July 2026

Studied fluid mechanics via case studies at the Panama Canal, a Penonomé wind farm, and Chiriquí hydroelectric facilities

Institut Notre-Dame de Bonne-Espérance

Braine-le-Comte, Belgium

Rotary Youth Exchange

June 2024

Study abroad program focused on cultural immersion and French language

COMPUTER & MACHINING SKILLS

CAD: SolidWorks, Autodesk Fusion, AutoCAD

Programming: C++, JavaScript, Python, MATLAB, Arduino

Manufacturing: 3D Printing (SLA, FDM, PµSL), CNC (3- & 5-axis), Manual Injection Molding, Vacuum Compression Molding, Rotary Axis Laser Cutting

Testing & Instrumentation: Mechanical Testing (Instron, ADMET), DMM Data Logging, Thermocouple Instrumentation

Electronics: PID Control Systems, Oscilloscopes, Soldering, Circuit Engraving (Laser Ablation)

PROFESSIONAL EXPERIENCE

MIT Lab for Translational Engineering

Cambridge, MA

Mechanical Engineering Co-op

January 2026 - Present

- Led mechanical design and fabrication across multiple concurrent device programs, taking each from CAD through prototyping and in-vivo validation
- Fabricated device hardware on SLA, PolyJet, and FDM printers, a 5-axis mill, and rotary-axis laser cutter
- Established repeatable fabrication and test protocols, training lab members to run them independently

FLX Solutions

Bethlehem, PA

Mechanical Engineering Intern

June - August 2025

- Collaborated with team to duplicate company's testbed for evaluating embedded software on hardware
- Programmed a custom application with JavaScript and the OpenAI API to automate company's task assignment protocol
- Trained a Microsoft TrOCR model in Python to convert handwritten notes into searchable database records

PROJECTS

Wetsuit Thermal Research | US Navy, MIT Lab for Translational Engineering

March 2026 - Present

- Engineered a high-pressure testing apparatus to evaluate the thermal conductivity of experimental diving materials at simulated depth, reproducing the skin-to-wetsuit interface with a PID-controlled heated surface
- Achieved pressures equivalent to 104 fsw and validated operation in both submerged and dry conditions, sealing heater and thermocouple feedthroughs against pressurized water with epoxy potting
- Synthesized novel materials and conducted mechanical stress testing to characterize structural integrity under high-pressure cycles

In-Situ Mixing Syringe | MIT Lab for Translational Engineering

February 2026 - Present

- Designed a dual-barrel syringe in Fusion 360 isolating two components until they mix at the tip, validated through a 23G needle
- Diagnosed leakage due to stress concentration at the gasket sealing interface and redistributed sealing load across 11 SLA-printed iterations

Stem-Cell Delivery Device | MIT Lab for Translational Engineering

February - June 2026

- Redesigned the central channel of a stem-cell delivery catheter, reducing taper and flow impedance to prevent cell shearing during in-vivo delivery
- Manufactured the complete device sets used for experiments, producing 40 assembled devices and 320 printed and machined parts

Vagus Nerve Stimulation PCB Housing | MIT Lab for Translational Engineering

February - April 2026

- Designed and fabricated a clear SLA housing contoured to the animal's back, sealed against fluid ingress at the wire ports and snap-fit lid seam
- Converged on assembly tolerances across 15 iterations, keeping the housing sealed and functional throughout in-vivo trials

Biomimetic Fan Blade Technology | Cornerstone of Engineering

January - March 2025

- Developed owl-wing-derived blade geometry in SolidWorks, reaching 30% noise reduction against a baseline design in flow simulation
- Tuned serration amplitude and wavelength against published aeroacoustic data after initial prototypes underperformed acoustically

INTERESTS

Classical Piano Performance, Astronomy, Tennis, Cross-Country Skiing, Hiking